Journal of Computational and Applied Mathematics 319 (2017) 514–520

Contents lists available at ScienceDirect

Journal of Computational and Applied Mathematics

journal homepage: www.elsevier.com/locate/cam

Adaptive cost dynamic time warping distance in time series analysis for classification

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HIGHLIGHTS

- Adaptive cost is introduced to dynamic time warping for better classification.
- Cost function is proposed for dynamic time warping.
- Experiments on UCR datasets prove that AC-DTW perform better than other methods.

ARTICLE INFO

Article history:
Received 4 September 2016
Received in revised form 20 October 2016

Keywords:
Time series classification
Dynamic time warping
Adaptive cost
Nearest neighbor classifier

ABSTRACT

Dynamic time warping (DTW) distance is commonly used in measuring similarity between time series for classification. In order to obtain the minimum cumulative distance, however, DTW distance may map multiple points on one time series to one point on another, and this makes time series over stretched and compressed, resulting in missing important feature information thus influence the classification accuracy. In this paper, we propose a method called adaptive cost dynamic time warping distance (AC-DTW), which adjusts the number of points on one time series mapped to the points on another. AC-DTW records the trajectories of all points and then adaptively allocates the cost rate to each point by calculating cost function at the next step. The results of the experiments implemented on 17 UCR datasets by using nearest neighbor classifier demonstrate that AC-DTW prevails in criterion of higher accuracy rate in comparison with some existing methods.

1. Introduction

The time series classification is an important topic in time series analysis and it has been applied in many areas, such as finance, biometrics, networking, artificial intelligence, etc. [1–4]. Given a time series sequence p, the goal of classification is to classify p in a dataset D into a prepared class. In this process, one of the important step is to calculate the distance between p and other time series in dataset D. To measure the distance between two time series, Euclidean distance and its variants are used most often. Agrawal et al. [5] proposed using Discrete Fourier Transform and Struzik et al. [6] proposed using Discrete Wavelet Transform. Although the improved Euclidean distance methods can translate the amplitude of time series, they failed to extend and warp the time axis. Dynamic time warping (DTW) is nowadays widely-used for measuring the distance, which was originally used in the field of text data matching and pattern recognition. Due to the elastic metric, it is introduced into the time series mining and classification by Berndt and Clifford [7], they use the dynamic programming [8] to align sequences with different lengths. While applying the DTW distance, the situation may occur that most points in a

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http://dx.doi.org/10.1016/j.cam.2017.01.004
0377-0427/

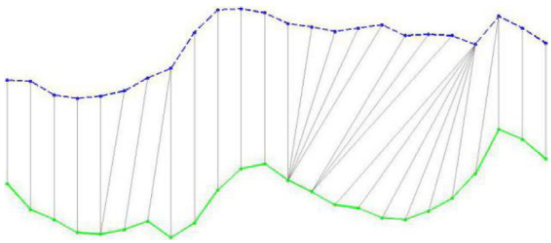

Fig. 1. The map of two time series under DTW distance.

time series are mapped to only a few points on another, and this makes the time series over stretched and compressed, leading to a singularity [9].

To avoid such extreme situation, the optimal warping path is limited to a search window, by the restriction method "Search window". Ratanamahatana and Keogh [10] proposed the adaptive window restriction method, knowing that the walking range of the optimum warping path is limited. It is most likely to miss the correct dynamic time warping distance in this case. Sakoe and Chiba [11] proposed slope constraint DTW, this method corrected the direction and step size of the optimal bending path, but it might destroy the continuity of the optimal warping path. Also, the selection of parameters is a difficult problem as we know already. Keogh and Pazzani [12] introduced the Derivative Dynamic Time Warping (DDTW), this method achieved a new sequence from the derivative of the original time series, then by applying the DTW distance on the new sequence, which can capture information about shape, but the step to find the new sequence will increase the computational complexity. Jeong et al. [13] proposed a weighted DTW (WDTW). Because of the DTW, it does not take relative significance depending on the phase difference between points into consideration. The proposed technique penalizes points with higher phase difference, between a reference point and a testing point to prevent minimum distance distortion resulted through outliers, but how to determine the weight depending on the different cases is difficult. Due to the high computational cost of DTW as the lengths of the signals increase, Barbon et al. [14] introduced an optimized version of the DTW that is based on the Discrete Wavelet Transform. And Salvador and Chan [15] proposed FastDTW, an approximation of DTW that has a linear time and space complexity.

In this paper, we propose an adaptive cost dynamic time warping distance (AC-DTW), which takes into consideration the excessive stretching or compression of the dynamic time warping distance. When choosing the optimal path, we limit the excessive tension or compression by giving the current step base distance an appropriate weight (greater than one), AC-DTW can avoid such an extreme situation that too many points in one sequence are mapped to only a few points in the other sequence.

The remainder of this paper is organized as follows. In Section 2, we introduce the background of the DTW method. In the following section, we describe the proposed method, which is called the AC-DTW, including the cost function, adaptive cost dynamic time warping distance, and together with the algorithm. Experiments and the performance are given in Section 4 and finally, we present our conclusion and the perspective of this work.

2. Original DTW

The dynamic time warping distance is now widely used in time series similarity measurement, which searches for the flexible corresponding relation between two time series. Fig. 1 gives a map of two time series under the DTW distance.

Consider two time series sequences $P = \{p_1, p_2, \dots, p_n\}$ and $Q = \{q_1, q_2, \dots, q_m\}$ of length n and m, we construct an $n \times m$ matrix, called distance matrix $D_{base} = (d_{base}(p_i, q_j))_{n\times m}$, where the entry $d_{base}(p_i, q_j)$ represents the base distance between points p_i and q_j. Then, we search an optimal warping path $R = \{r_1, \dots, r_k, \dots, r_K\}$, which minimizes the accumulation of corresponding distance, where $\max(m, n) \le K < m + n - 1$, Fig. 2 shows the optimal warping path between P and Q in the distance matrix.

The warping path is typically subjected to the following constraints.

(1) Boundary conditions: the first and the last points in one time series should map to the other series, i.e. $r_1 = D(1, 1)$, $r_K = D(n, m)$;
(2) Monotonicity: for $r_k = D(i_k, j_k)$ and $r_{k+1} = D(i_{k+1}, j_{k+1})$, $i_{k+1} \ge i_k$ and $j_{k+1} \ge j_k$;
(3) Continuity: $i_{k+1} \le i_k + 1$ and $j_{k+1} \le j_k + 1$.

So, the dynamic time warping distance is obtained by the minimum warping path distance. Therefore, the optimal warping path formula can be described as follows

$$DTW(P, Q) = \min_W \left[\sum_{k=1}^{K} d_{base}(r_k)\right] \quad (1)$$

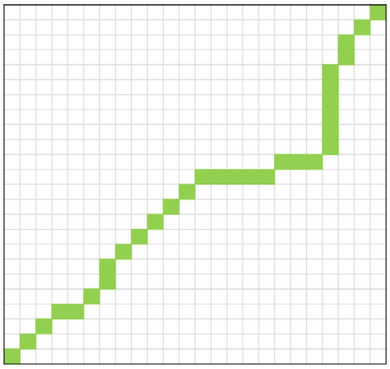

Fig. 2. The optimal warping path in distance matrix.

where $d_{base}(w_k) = d_{ij} = d_{base}(p_i, q_j) = (p_i - q_j)^2$ represents the distance of mapped points in the warping path. This path, can be efficiently founded by using dynamic programming [1]. To evaluate Eq. (1) we follow

$$DTW(i, j) = d_{base}(p_i, q_j) + \min \begin{cases} DTW(i, j-1) \\ DTW(i-1, j-1) \\ DTW(i-1, j). \end{cases} \quad (2)$$

Finally we can obtain the matrix DTW, where DTW$(1, 1) = d_{base}(p_1, q_1)$, and DTW$(n, m)$ is the DTW distance.

We assume, that there are two time series $S_1 = [2, 1, 2, 1]$ and $S_2 = [1, 2, 1]$, the first step is to compute the base distance matrix between the two time series,

$$D = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

Then compute the matrix DTW by dynamic programming.

$$\mathrm{DTW} = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \end{bmatrix}.$$

Thus the minimum warping distance is given by $D(3, 4) = 1$.

3. Adaptive cost dynamic time warping

To avoid the extreme situation that too many points in one sequence are mapped to only a few points on the other one, some dynamic time warping distance variants have been proposed [16]. A typical one is DDTW [12], derivative DTW, which captures information about shape by applying DTW distance on the new sequence. It starts from the derivative of the original time series, but the procedure of finding this new sequence increases computational complexity. AC-DTW, proposed in this paper, differently uses matrices constantly record the number of points mapping to each point in the original path, when finding the optimal warping path, the cumulative distance is no longer based on the base distance, but on the product of the base distance and a positive number related to the number of mapping points, also the positive number is the adaptive cost defined in this paper. In this section, the cost function will be introduced, and then AC-DTW and its algorithm.

3.1. Cost function

We use matrices to constantly record the number of points mapping to each point in the current path, when we find the optimal warping path. Firstly, we define two matrices $A = (a_{ij})_{n\times m}$ and $B = (b_{ij})_{n\times m}$, where the entries record the times of being used of each point in time series P and Q. Their initial values are null matrix, and in the optimal warping path AC-DTW(s, t), the point p_s is used $a_{s,t}$ times, and the point q_t is used $b_{s,t}$ times.

After that, we define a cost function in (3), which adjusts the frequency of each point, and we add the cost function to the base distance as a weight when a point is repeatedly used.

$$c(x) = g \cdot r \cdot x + 1. \quad (3)$$

In cost function (3), g is a positive parameter, which is used to adjust the effect of cost function $c(x)$, higher value of g amplifies the effect of $c(x)$ while lower value diminishes the effect. The term x denotes how many times that each points are used in time series, thus it is obvious that the cost function is positively proportional to x, meaning that when x is greater, so is the cost function simultaneously. Then when the adjusted distance between mapped points is higher from (6) in the

Table 1
The pseudo code of AC-DTW algorithm.

```
Input: time series P={p1,p2,...,pn} and Q={q1,q2,...,qm}, parameter g, r=min(n,m)/max(n,m), base distance matrix Dbase;
Output: the AC-DTW distance between two time series AC-DTW(n,m), matrix Path contain the warping path;
Initialize: A=0, B=0, AC-DTW(1,1)=d11; A(1,1)=1; B(1,1)=1
for i=1:n
    for j=1:m
      d1=Dbase(i,j)*(g*r*B(i-1,j)+1) + AC-DTW(i-1,j);
      d2=Dbase(i,j) + AC-DTW(i-1,j-1);
      d3=Dbase(i,j)*(g*r*A(i,j-1)+1) + AC-DTW(i,j-1);
      AC-DTW(i,j)=min(d1,d2,d3);
      Path_i,j=min_index((i-1,j),(i-1,j-1),(i,j-1))
       if d1== AC-DTW(i,j)
        A(i,j)=1; B(i,j)=B(i-1,j)+1;
      else if d2== AC-DTW(i,j)
          A(i,j)=1; B(i,j)=1;
        else A(i,j)=A(i,j-1)+1; B(i,j)=1;
      end if
    end for
end for
```

next section, the situation of many-to-one is more unacceptable, and vice versa. However, if the lengths of P and Q are very different, the situation of many-to-one gets more popular, thus we add the term r, which is

$$r = \min(n, m)/\max(n, m). \tag{4}$$

It is used to numerically control the endurance of many-to-one situation. It is obvious that the greater difference between m and n is, the closer the value of r is to 1, and particularly,

$$r = \begin{cases} = 1, & m = n \\ < 1, & m \neq n \end{cases} \tag{5}$$

the value of (5) tends to 1 when the length of one time series is close to another, and the value of the cost function also increases with this trend, indicating that when lengths of the two time series are closer, the many-to-one situation is more unacceptable.

3.2. Adaptive cost dynamic time warping distance

To avoid the extreme situation that too many points on one sequence are mapped to one point on another, we define a cost function (3), as weight to the base distance to determine the next step, based on the continuity and monotonicity of AC-DTM, and then we solve the adaptive cost dynamic time warping distance with the following dynamic method:

$$AC\text{-}DTW(i,j) = \min \begin{cases} c(b_{i-1,j}) \times d_{i,j} + AC\text{-}DTW(i-1,j) & \text{(i)} \\ d_{i,j} + AC\text{-}DTW(i-1,j-1) & \text{(ii)} \\ c(a_{i,j-1}) \times d_{i,j} + AC\text{-}DTW(i,j-1) & \text{(iii).} \end{cases} \tag{6}$$

Case (i) shows that the warping path is from AC-DTW $(i-1, j)$ to AC-DTW(i, j), meaning that point q_j is reused, then we adjust the many-to-one case by imposing the cost $c(b_{i-1,j})$ on the current base distance d_{ij}, meanwhile we set $a_{i,j} = 1$ and $b_{i,j} = b_{i-1,j} + 1$.

In case (ii), the warping path is from AC-DTW$(i-1, j-1)$ to AC-DTW(i, j), meaning that there is no reused point, then no cost is imposed on and we set $a_{i,j} = 1$ and $b_{i,j} = 1$.

In case (iii), the warping path is from AC-DTW$(i, j-1)$, meaning that p_i is reused, then we impose the cost $c(a_{i,j-1})$ on the current base distance d_{ij}, and set $a_{i,j} = a_{i,j-1} + 1$ and $b_{i,j} = 1$.

Therefore according to formula (6), if a point is reused, the current base distance d_{ij} will be added up to a larger value, which leads to the abandon of this point via minimization. Thus this adaptive cost dynamic time warping controls the extreme situation of many-to-one.

3.3. Algorithm

The pseudo code of AC-DTW algorithm is summarized in Table 1, where $D_{base} = (d_{ij})_{n\times m}$ is the pre-calculated Euclidean distance matrix between P and Q, and Path is a matrix, which contains the warping path. Finally we can get the AC-DTW distance AC-DTW(n, m) between two time series and the warping path.

Table 2
Summary of datasets.

Datasets	Classes	Size of training set	Size of testing set	Time series length
ECG200	2	100	100	96
Gun_Point	2	50	150	150
FISH	7	175	175	463
CBF	3	30	900	128
OSULeaf	6	200	242	427
Trace	4	100	100	275
FaceFour	4	24	88	350
FaceAll	14	560	1690	131
Beef	5	30	30	470
Oliveoil	4	30	30	570
Adiac	37	390	391	176
Lighting2	2	60	61	637
Coffee	2	28	28	286
Cricket_X	12	390	390	300
Cricket_Y	12	390	390	300
Cricket_z	12	390	390	300
Symbols	6	25	995	398

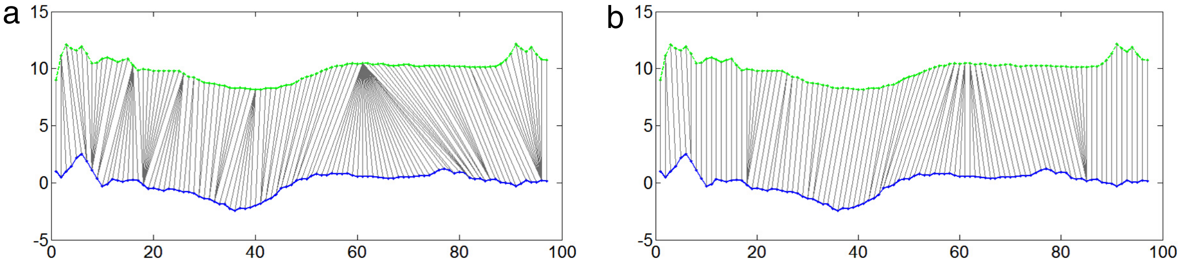

Fig. 3. Graphs of correspondence between two series of ECG200 dataset based on DTW (a) and AC-DTW (b). In the measure of DTW, there are 29 points map to one point, but in the measure of AC-DTW there are at most 6 points map to one.

4. Experiments

In this section, we implement experiments on 17 datasets for classification to verify the performance of AC-DTW distance by comparing different measurement methods. We use nearest neighbor as classifier. Description of data, experimental results and the time complexity are discussed in this section.

4.1. Data sources

The test data of this paper is from "UCR Time Series Data Mining Archive" [17]. In this paper, we choose 17 datasets to test the accuracy and robustness of AC-DTW, which is shown in Table 2, different numbers of time series, different lengths of time series, and different numbers of classes are selected.

4.2. Experimental settings

Algorithm in this paper is programmed with C++ language, and as for the classification process, the nearest neighbor classifier is chosen. Different distance measurements, such as ED (Euclidean Distance), DTW, BDTW (DTW of Best Warping Window) and AC-DTW, are chosen for comparison. And algorithms are evaluated based on their classification accuracy (ACC) in 17 datasets, parameter $g = 0.5$ due to the former comparison and ACC is defined as

$$\mathrm{ACC} = \frac{\text{Number of time series being correctly classified}}{\text{Total number of time series of test set}} \tag{7}$$

and then we calculate misclassification rate (MR) $= 1 - \mathrm{ACC}$, which is the criterion of the comparison.

4.3. Experimental results

4.3.1. Correspondence between two series

For comparative analysis, we first display the correspondence relationship between two time series based on original DTW and AC-DTW, shown in Fig. 3. These two series are the No.4 and No.11 series of the ECG200 data set in Time Series

Table 3
Misclassification rate (MR) of different measure methods on different datasets. (The best results are highlighted in bold.)

Datasets	ED	DTW	BDTW	AC-DTW(0.5)-fivefold
ECG200	**0.12**	0.23	**0.12**(**0**)	0.17 ± 0.01
Gun_Point	0.087	0.093	0.087(0)	**0.025 ± 0.02**
FISH	0.217	0.167	0.16(4)	**0.148 ± 0.03**
CBF	0.148	0.003	0.004(11)	**0**
OSULeaf	0.483	0.409	0.384(7)	**0.307 ± 0.07**
Trace	0.24	**0**	0.01(3)	**0**
FaceFour	0.216	0.17	**0.114**(**2**)	**0.06 ± 0.02**
FaceAll	0.286	0.192	0.192(3)	**0.177 ± 0.03**
Beef	0.467	0.5	0.467(0)	**0.312 ± 0.06**
Oliveoil	**0.133**	**0.133**	0.167(1)	0.15 ± 0.06
Adiac	0.389	0.396	0.391(3)	**0.35 ± 0.02**
Lighting2	0.246	0.131	0.131(6)	**0.112 ± 0.06**
Coffee	0.25	0.179	0.179(3)	**0**
Cricket_X	0.426	0.223	0.236(7)	**0.175 ± 0.03**
Cricket_Y	0.356	0.208	**0.197**(**17**)	0.212 ± 0.03
Cricket_z	0.38	0.208	0.18(7)	**0.174 ± 0.02**
Symbols	0.1	0.05	0.062(8)	**0.017 ± 0.008**

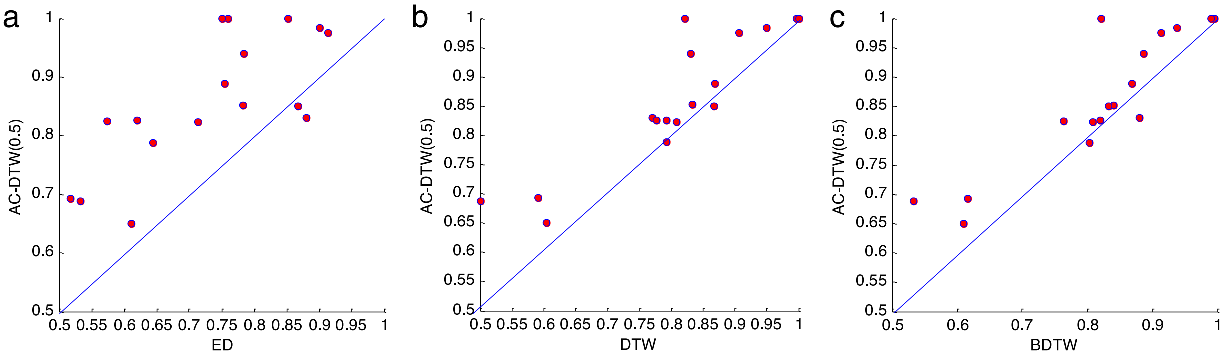

Fig. 4. Graphical comparison of accuracy rates from different methods. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Archive UCR. Fig. 3(a) is alignment of sequences based on original DTW, from which we can see that multiple points in the top time series has been mapped to one point of the time series below, and the extreme case is that 29 points map to one point above, which means that the top time series is severely compressed, and features on this segment is missed in the process of compression. Fig. 3(b) is based on method AC-DTW of this paper, and we can clearly see that 6 points at most on one time series has mapped to one point on the other. This result greatly improves the compression of the time series in comparison with the original dynamic time warping. Thus, more features of the time series are captured, which illustrates the validity of our method.

4.3.2. Misclassification rate values

In order to make our experimental results more convincing, we mix the test data and the training data as a new data set, then we use 5-fold cross-validation for the proposed method on the new data. And the average misclassification rate (MR) values are chosen as the criteria for comparison among different methods. The experimental results are summarized in Table 3, we can see that AC-DTW performs the best among all the compared methods on 14 datasets out of 17, and it significantly reduces the misclassification rate especially for OSULeaf, Face Four, Beef and Coffee.

4.3.3. Graphical comparison of accuracy

Graphical demonstration of comparison between AC-DTW(0.5) and other methods of DTW are given in Fig. 4. On each graph, there are 17 points representing the quotients of accuracies of two different methods on 17 datasets and the straight blue line stands for the identical line. Thus points above the blue lines in the figure show that the accuracy of AC-DTW(0.5) is greater, otherwise points below the blue line mean the compared method is better.

Fig. 4(a) shows very clearly that the accuracy of AC-DTW is much greater than that of Euclidean distance on 15 datasets, of which the ratio reach 88.23%; Fig. 4(b) and (c) also show higher accuracy of AC-DTW than DTW on 14 datasets and than DTW (best warping window) on 15 datasets as well, of which the ratios reach 82.35% and 88.23% respectively.

According to the experimental results analysis above, time series classification accuracy of method AC-DTW of this paper is greatly improved by introducing the cost function, being compared with several state-of-the-art methods.

4.4. Time complexity

To compute the AC-DTW distance of two time series with the length n and m, we need to search for the optimal warping path on an $n \times m$ matrix, then the time complexity is $O(nm)$, which is the same as DTW. Then we use N_1 time series in length m for training, classify N_2 time series in length n, and compute the classification accuracy, therefore the time complexity is $O(N_1N_2nm)$.

5. Conclusions

In this paper we propose an adaptive cost dynamic time warping (AC-DTW) algorithm to compute the similarity of two time series. AC-DTW may avoid the drawback that too many points on one sequence are aligned to one point on another. We introduce a cost function $C(x)$ as a weight imposed on current distance and then use it to define an adaptive cost dynamic time warping distance, AC-DTW. AC-DTW does not increase the computational complexity, and capture more property of the time series. Experimental results also demonstrate that the proposed method outperformed several existing methods. The future work will be focused on the selection of parameter g and the combination of AC-DTW with other dynamic time warping distance variants.

Acknowledgments

This work is supported by Major projects of the National Natural Science Foundation of China (91324201), which titled Rules and Mechanisms of Psychological and Behavioral Changes of Social Groups in Unconventional Emergencies and Natural Science Foundation of China (81271513), which titled Research on Prewarning and Intervening of Psychological Crisis Based on Structural Equation Models and Effect Analysis. And we also gratefully acknowledge Yanping Chen, Eamonn Keogh, Bing Hu, et al., for their devoted work on the collection of UCR datasets which support our work.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at http://dx.doi.org/10.1016/j.cam.2017.01.004.

References

[1] E. Keogh, S. Kasetty, On the need for time series data mining benchmarks: A survey and empirical demonstration, Data Min. Knowl. Discov. 7 (4) (2003) 349–371.
[2] B. Legrand, C.S. Chang, S.H. Ong, et al., Chromosome classification using dynamic time warping, Pattern Recognit. Lett. 29 (3) (2008) 215–222.
[3] T.M. Rath, R. Manmatha, Word image matching using dynamic time warping, in: 2003 IEEE Computer Society Conference on Computer Vision and Pattern Recognition. Proceedings, vol. 2, IEEE, 2003, pp. 521–527.
[4] M. Bashir, J. Kempf, Reduced dynamic time warping for handwriting recognition based on multidimensional time series of a novel pen device, Int. J. Intell. Syst. Technol. (WASET) 3 (4) (2008) 1839–1845.
[5] R. Agrawal, C. Faloutsos, A. Swami, Efficient similarity search in sequence databases, in: Proc.of Intl Conf.on Data Organazation, 730, 1993, pp. 69–84.
[6] Z.R. Struzik, A. Siebes, Wavelet transform in similarity paradigm, in: Proceedings of the Second Pacific-Asia Conference on Research and Development in Knowledge Discovery and Data Mining. 1998, pp. 295–309.
[7] D.J. Berndt, J. Clifford, Using Dynamic Time Warping to Find Patterns in Time Series, KDD workshop. 10 (16) (1994) 359–370.
[8] D.J. Berndt, J. Clifford, Finding patterns in time series: a dynamic programming approach, in: Advances in Knowledge Discovery and Data Mining, American Association for Artificial Intelligence, 1996, pp. 229–248.
[9] D. Yu, X. Yu, Q. Hu, et al., Dynamic time warping constraint learning for large margin nearest neighbor classification, Inform. Sci. 181 (13) (2011) 2787–2796.
[10] V. Niennattrakul, C.A. Ratanamahatana, On clustering multimedia time series data using K-Means and dynamic time warping. in: Proceedings of the International Conference on Multimedia and Ubiquitous Engineering. 2007, pp. 733–738.
[11] H. Sakoe, S. Chiba, Dynamic programming algorithm optimization for spoken word recognition, Trans. Acoust. Speech Signal Process. 1 (26) (1978) 43–49.
[12] E.J. Keogh, M.J. Pazzani, Derivative dynamic time warping, in: Proceedings of the 2001 SIAM International Conference on Data Mining. 2001,pp. 1–11.
[13] Y.S. Jeong, M.K. Jeong, O.A. Omitaomu, Weighted dynamic time warping for time series classification, Pattern Recognit. 44 (9) (2011) 2231–2240.
[14] S. Barbon, R.C. Guido, L.S. Vieira, et al., Wavelet-based dynamic time warping, J. Comput. Appl. Math. 227 (2) (2009) 271–287.
[15] S. Salvador, P. Chan, Toward accurate dynamic time warping in linear time and space, Intell. Data Anal. 11 (5) (2007) 561–580.
[16] Y.S. Jeong, R. Jayaraman, Support vector-based algorithms with weighted dynamic time warping kernel function for time series classification, Knowl.-Based Syst. 75 (2015) 184–191.
[17] Yanping Chen, Eamonn Keogh, Bing Hu, et al. The UCR Time Series Classification Archive. 2015. URL www.cs.ucr.edu/~eamonn/time_series_data/.